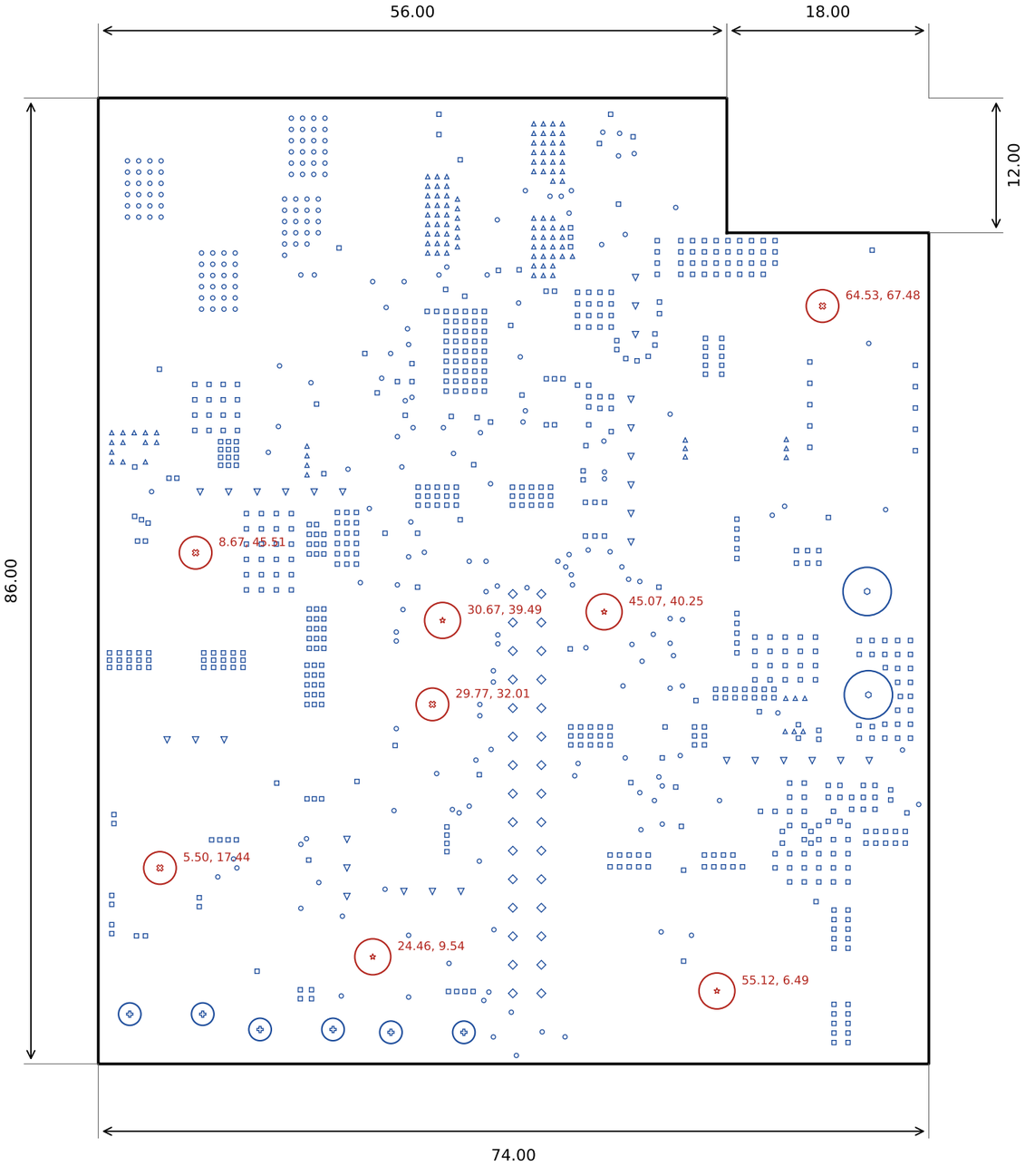


Datum: lower-left corner of the outline (0, 0). Dimensions in mm, outline tolerance +/-0.2 mm.
Drawn from the Excellon files: every drilled hole is shown with its symbol; holes >= 2 mm also to scale.

BOOST POWER BOARD - FABRICATION DRAWING

BOOST_power_RC2, 2026-10-13. Files: BOOST_power_RC2_gerbers.zip. Units: mm.

FABRICATION NOTES

1. **ALL VIAS EPOXY FILLED AND COPPER CAPPED (resin plugged, plated over; IPC-4761 Type VII). 468 of the 972 vias sit in SMD pads. Via drills 0.30, 0.40, 0.50 mm (JLCPCB filled-via range 0.2-0.5 mm). Order option "Via Covering: Epoxy Filled & Capped".**
2. Material FR-4, Tg 155 C or better (JLCPCB "FR4 TG155"). 6 copper layers. Finished thickness 1.6 mm +/-10 %.
3. Copper 1 oz (35 um) finished on all six layers: outer 1 oz, inner 1 oz (order option "Inner copper weight 1 oz"; the fab default of 0.5 oz inner is NOT acceptable).
4. **STACKUP JLC061611-7628D, as in the table (1.583 mm nominal; order options "Specify Stackup: Yes" and "JLC061611-7628D"). Not the default build: the layer spacing sets the switching-loop inductance. No impedance control.**
5. Surface finish ENIG. Solder mask green LPI both sides. Silkscreen white both sides.
6. Minimum track / space: 0.20 / 0.20 mm on every layer. Minimum drill 0.30 mm (vias); smallest plated pad hole 1.05 mm.
7. Minimum via annular ring 0.10 mm; minimum plated pad annular ring 0.20 mm. Via-to-via hole gap >= 0.25 mm; filled via to any pad hole >= 0.48 mm.
8. Plated hole tolerance +0.13 / -0.08 mm. J10 2x15 header holes are drilled 1.05 mm for Samtec 0.64 mm square posts.
9. Non-plated holes: 4 x 3.20 mm (standoffs), 4 x 2.90 mm (FET tab screws).
10. Outline on the Edge.Cuts (GM1) layer, tolerance +/-0.2 mm. Notched top-right corner.
11. Electrical test: 100 % flying probe; IPC-D-356 netlist BOOST_power_RC2.ipc356 supplied.
12. No order number or other marking to be added ("Remove Mark").

STACKUP (from the board file; copper + dielectric 1.583 mm, + mask)

F.Mask	solder mask		0.010
F.Cu	copper, 1 oz	signal, parts, pours	0.035
	prepreg 2 x 7628, er 4.4		0.421
In1.Cu	copper, 1 oz	GND plane	0.030
	core, er 4.6		0.130
In2.Cu	copper, 1 oz	LX / Vin pours, signal	0.030
	prepreg 3 x 3313, er 4.1		0.291
In3.Cu	copper, 1 oz	Vin/Vout pours, signal	0.030
	core, er 4.6		0.130
In4.Cu	copper, 1 oz	Vin/Vout pours, signal	0.030
	prepreg 2 x 7628, er 4.4		0.421
B.Cu	copper, 1 oz	signal, parts, pours	0.035
B.Mask	solder mask		0.010

DRILL TABLE

sym	drill mm	type	count	used for
○	0.30	PTH	231	vias (filled, capped)
□	0.40	PTH	627	vias (filled, capped)
△	0.50	PTH	114	vias (filled, capped)
◇	1.05	PTH	30	2x15 socket (J10)
▽	1.10	PTH	30	TO-220 pins M1-M10
✚	2.00	PTH	6	LED wire pads J3-J8
⊠	2.90	NPTH	4	FET tab screws
☆	3.20	NPTH	4	standoffs
⬡	4.30	PTH	2	battery lugs J1/J2